

Predicting Fair Pricing of Wedding Photography Vendors Using Web-Scraped Vendor Data

Business Analytics Individual Case Study

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1. Problem Statement and Objectives

1.1 Problem Statement

Wedding photography is a high-value service in which customers often have limited information about whether a quoted price is reasonable relative to comparable vendors. Unlike standardized products, photography services differ across vendors in terms of location, reputation, review history, experience, service packages and platform-level trust indicators.

This creates an information gap between vendors and customers. A customer may observe substantially different prices for apparently similar photography services but have no objective reference point for determining whether a particular quote is relatively low, typical or high for its market and observable vendor characteristics.

This case study addresses the following business question:

Can publicly observable characteristics of wedding photography vendors be used to estimate an expected market price and classify vendors as Underpriced, Fairly Priced, or Overpriced relative to that benchmark?

The study uses web-scraped vendor listings to construct a data-driven pricing benchmark and subsequently develops a classification model using information that could be observed before booking.

The terms **Underpriced**, **Fairly Priced**, and **Overpriced** in this study are therefore **relative analytical categories**, not claims about the intrinsic quality or objective value of an individual vendor.

1.2 Objectives

The study has five primary objectives:

1. **Data Collection:** Build a self-collected dataset of at least 10,000 wedding photography vendor listings using publicly available web information.
2. **Data Preparation:** Clean the collected data while preserving genuine missing values and avoiding fabricated vendor attributes.
3. **Price Benchmarking:** Estimate an expected vendor price using observable market, reputation and service characteristics through a hedonic pricing approach.
4. **Price Classification:** Use out-of-fold pricing deviations to classify vendors into Underpriced, Fairly Priced and Overpriced relative to the estimated benchmark.
5. **Predictive Analytics:** Determine whether pre-booking characteristics can predict these pricing categories and derive practical insights for customers, vendors and marketplace platforms.

2. Dataset Source and Collection Method

2.1 Data Source

The dataset was self-collected from **WedMeGood**, a wedding marketplace containing publicly accessible wedding vendor listings.

The collection focused on wedding photography vendors across **14 Indian metropolitan and destination markets**.

The raw dataset contained:

- **11,340 vendor listings**
- **19 raw attributes**
- Vendor identification information

- City and locality
- Starting daily price
- Small-event price where available
- Rating
- Review count
- Package information
- Videography availability
- Number of cities covered
- Platform badges
- Experience information
- Source URL and collection timestamp

The dataset was therefore created specifically for this case study rather than obtained from Kaggle, UCI or an existing GitHub dataset.

2.2 Data Collection Method

A custom Python web-scraping process was used to collect publicly accessible vendor listing information.

The collected attributes included:

Category	Examples
Vendor information	Vendor ID, vendor name
Location	City, base city, locality
Pricing	Starting daily price, small-event price
Reputation	Rating, review count
Experience	Stated experience in vendor biography
Services	Package type, videography
Market reach	Cities covered
Platform signals	Award, handpicked and verified status
Audit information	Source URL, source name, scraping timestamp

The collection process was restricted to publicly accessible commercial listings. No private customer information was collected.

The project also preserved the source URL associated with each listing to maintain traceability of the collected observations.

2.3 Dataset Size

The raw dataset contained **11,340 records**.

After data preparation:

- 160 duplicate vendor listings were removed.
- 147 records without an observed starting daily price were removed.
- 5 extreme price observations outside the defined commercial range were removed.

The final analytical dataset contained:

11,030 vendor listings

This exceeds the minimum requirement of 10,000 records.

3. Data Preparation and Exploratory Analysis

3.1 Data Cleaning

Data preparation was performed before modelling to ensure that the analysis did not introduce artificial information.

Price

The primary pricing variable was:

price_per_day_inr

This represents the directly observed starting daily rate from the listing.

No artificial minimum or maximum price ranges were generated.

In particular, no formulas such as:

- $0.7 \times \text{observed price}$
- $1.5 \times \text{observed price}$

were used to manufacture missing price ranges.

Records without an observed starting daily price were excluded because price is fundamental to the benchmark being constructed.

3.2 Duplicate Handling

Vendor IDs were used to identify duplicate listings.

A total of **160 duplicate vendor listings** were removed.

The final dataset therefore contains unique vendor identifiers for the analytical sample.

3.3 Rating and Review Treatment

A substantial proportion of vendors had no rating/review information.

The final dataset contains:

- **5,040 rated vendors**
- **5,990 unrated vendors**

Rather than treating an unrated vendor as a low-rated vendor, the raw rating value of zero was retained and an `is_rated` indicator was created.

For continuous modelling, the rating value was imputed only where required, while the rating-availability indicator was retained separately.

This distinction is important because:

No rating is not equivalent to a poor rating.

3.4 Experience

Experience was directly extracted only where it was explicitly available in the vendor information.

The data contained:

- **892 vendors with directly stated experience**
- **10,138 vendors without stated experience**

The unstated experience values were not inferred from review counts.

For modelling purposes, missing experience was median-imputed to **8 years**, with an additional `experience_stated` indicator identifying whether the original experience was actually observed.

A sensitivity analysis was subsequently performed to examine whether this modelling decision materially affected classification performance.

3.5 Locality and Other Categorical Variables

Missing locality values were preserved as **Unknown** rather than assigning artificial locations.

Similarly, package categories were standardized without inventing characteristics that were not present in the source listings.

These decisions were made to ensure that data preparation did not create information that was absent from the original web source.

3.6 Exploratory Data Analysis

The final dataset was examined using descriptive statistics and visualizations.

Price Distribution

Wedding photography prices displayed substantial right skew.

The final dataset had approximately:

- **Mean price:** ₹66,312
- **Median price:** ₹50,000
- **Interquartile range:** ₹30,000–₹80,000

The strong difference between the mean and median indicates that a relatively small number of high-priced listings increase the average.

A logarithmic transformation of price was therefore used for the pricing benchmark.

3.7 Geographic Differences

The dataset showed substantial variation across markets.

The analysis identified higher median prices in markets such as:

- Bangalore
- Delhi NCR
- Mumbai
- Udaipur

while lower median prices were observed in markets such as:

- Kolkata
- Surat
- Patna

This indicates that location is an important component of price variation.

However, these differences should be interpreted as **market-level associations**, since the dataset is observational and does not establish that location itself causes the difference.

3.8 Reviews and Ratings

The analysis found that review volume had a stronger relationship with pricing than the star-rating scale alone.

This is plausible because star ratings were highly concentrated toward the upper end among rated vendors, limiting their ability to distinguish vendors.

Review volume provided greater variation between listings.

The Random Forest analysis later identified review count as the largest individual feature importance at approximately **23.6%**.

This suggests that the amount of accumulated social proof was an important predictive signal in the classification model.

3.9 Experience and Platform Signals

Vendors with explicitly stated experience showed different pricing patterns from vendors without stated experience.

Platform-level attributes such as awards, handpicked status and verification were also examined as potential explanatory variables.

However, these variables should be interpreted as **observable marketplace signals**, rather than direct measures of photographic quality.

4. Analytics Method and Implementation

4.1 Analytical Framework

The study uses a two-stage analytical framework.

Stage 1 — Expected Price Benchmark

A hedonic pricing model estimates the expected price of a vendor using observable characteristics.

Stage 2 — Pricing Classification

The difference between the actual price and the expected price is used to classify each vendor relative to the benchmark.

A third stage then tests whether the resulting pricing categories can be predicted using pre-booking vendor characteristics.

The overall workflow is:

Web Data → Cleaning → EDA → Expected Price Model → OOF Residuals → Pricing Classes → Classification Model → Business Insights

4.2 Hedonic Pricing Model

The study follows the general logic of the **hedonic pricing approach**, where the price of a differentiated service can be related to its observable characteristics.

The model estimates:

$$\ln(\text{Price}_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \epsilon_i$$

where:

- $Price_i$ = observed starting daily price
- X_i = observable vendor characteristics
- ϵ_i = pricing deviation not explained by the included variables

A Ridge regression model with:

$$\alpha = 1.0$$

was used to reduce instability caused by correlated predictors.

The model incorporated market, reputation, experience, service and platform-related characteristics.

4.3 Out-of-Fold Validation

A major methodological consideration was preventing the price benchmark from being overly optimistic.

If the expected price were calculated from a model trained on the same observation, the resulting residual could be artificially small.

Therefore, the study used **5-fold cross-validation with shuffled folds**.

For every vendor:

1. The model was trained using the other folds.
2. The held-out vendor's expected log price was predicted.
3. The observed log price was compared with this out-of-fold prediction.
4. The resulting difference became the vendor's OOF residual.

The residual was calculated as:

$$e_i = \ln(Price_i) - \widehat{\ln(Price_i)}_{OOF}$$

This ensures that each vendor's benchmark deviation is calculated using a model that did not train on that vendor.

The mean OOF R^2 was:

0.0869

The relatively modest explanatory power is important: the available web attributes explain only a portion of observed price variation. This is expected in a service such as wedding photography, where many relevant factors—such as artistic style, negotiation, exact deliverables and client-specific requirements—may not be available in standardized listing data.

4.4 Fair-Price Classification

Instead of choosing an arbitrary rule such as $\pm 20\%$, the study uses the empirical distribution of OOF residuals.

The observed boundaries were:

$$Q1 = -0.3363 \quad Q_1 = -0.3363$$

and

$$Q3 = +0.2973 \quad Q_3 = +0.2973$$

The resulting categories were:

Category	Rule	Vendors	Share
Underpriced relative to benchmark	Residual < -0.3363	2,758	25.0%
Fairly Priced relative to benchmark	Q1 to Q3	5,514	50.0%
Overpriced relative to benchmark	Residual > +0.2973	2,758	25.0%

The lower boundary corresponds approximately to a price more than **28.6% below** the model's expected level, while the upper boundary corresponds approximately to a price more than **34.6% above** the expected level.

These categories are **benchmark-relative classifications**. They do not imply that a vendor's actual service quality is objectively low or high.

4.5 Predictive Classification

The next objective was to determine whether pricing categories could be predicted using information available before a customer books a vendor.

Two classification models were evaluated:

1. Multinomial Logistic Regression
2. Random Forest Classifier

The dataset was divided using an **80:20 stratified train-test split**:

- Training: **8,824 records**
- Testing: **2,206 records**

Price and price-derived variables were excluded from the classifier.

The excluded variables included:

- actual price
- log price
- OOF expected price
- OOF residual
- price deviation

This prevented target leakage.

4.6 Model Results

Model	Test Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Random Guessing	33.33%	33.33%	33.33%	0.3333	0.3333

Logistic Regression	39.17%	36.19%	37.58%	0.3537	0.3977
Random Forest	41.12%	37.56%	38.31%	0.3657	0.4100

The Random Forest achieved the highest test accuracy and Macro F1 among the evaluated models.

Its accuracy was approximately **7.79 percentage points above random guessing**, equivalent to a relative improvement of approximately 23.4% over the random baseline.

However, the overall accuracy remains modest. This is an important result rather than something to hide: publicly observable listing characteristics provide useful but incomplete information for predicting relative pricing categories.

4.7 Feature Importance

The Random Forest model identified the following major predictive signals:

- Review count: approximately **23.6%**
- Market/city effects: approximately **18.6%**
- Customer rating: approximately **15.0%**
- Operating tenure: approximately **13.4%**
- Cities covered: approximately **11.5%**

The importance values indicate predictive contribution within the fitted Random Forest.

They **do not establish causality**.

For example, the finding that review count is important does not prove that obtaining additional reviews directly causes a vendor to increase prices.

4.8 Experience Sensitivity Analysis

Because experience was unstated for approximately 91.9% of vendors, an additional sensitivity analysis was conducted.

Model A

Included:

- median-imputed experience
- experience-stated indicator

Results:

- Accuracy: **41.12%**
- Macro F1: **0.3657**

Model B

Removed the continuous imputed experience variable while retaining the experience-stated indicator.

Results:

- Accuracy: **41.61%**
- Macro F1: **0.3718**

The difference was small:

- Accuracy: +0.49 percentage points
- Macro F1: +0.0061

This indicates that the continuous imputed experience variable has **limited influence on the observed classification performance**. The result supports retaining the disclosure indicator while treating the imputed continuous experience variable cautiously.

5. Comparison with Three Published Studies

Recent research provides useful context for the present analysis.

5.1 Carnehl, Stenzel and Schmidt

Carnehl, Stenzel and Schmidt's *Management Science* study examines dynamic pricing in the presence of rating systems. The authors analyze how pricing and ratings can interact on online platforms and show that the relationship between price and reviews can involve strategic feedback mechanisms.

This is relevant to the present study because ratings and reviews are important observable signals in online marketplaces.

Comparison

The present study differs in several ways:

- It focuses on wedding photography rather than a general online-platform setting.
- It uses vendor-level web-scraped listings.
- It examines review volume and rating availability as predictors of benchmark-relative pricing categories.
- It does not attempt to model strategic dynamic pricing behaviour.

Thus, the published work provides theoretical motivation for considering platform reputation signals, while the present study applies those signals in an empirical service-pricing benchmark.

5.2 Namburu, Selvaraj and Varsha

Namburu, Selvaraj and Varsha investigate machine-learning approaches to competitive product pricing using reviews and statistical/categorical features. Their published work proposes a hybrid X-NGBoost approach and compares it with other ensemble boosting methods.

Comparison

The present study shares the use of marketplace attributes and machine learning for pricing-related analysis.

However, there are important differences.

The current study:

- focuses on a service rather than standardized products
- uses wedding photography vendors
- constructs an expected-price benchmark before classification
- uses Random Forest and Logistic Regression for pricing-tier classification
- emphasizes interpretability and benchmark-relative categories rather than maximizing continuous price-prediction accuracy

The comparison therefore supports the broader use of machine learning for marketplace pricing while highlighting the importance of domain-specific modelling.

5.3 Tan et al.

Tan et al. examine Airbnb rental price prediction using multimodal data, including structured attributes, text and images. Their study uses 8,380 Airbnb listings from Amsterdam and develops a multimodal deep-learning approach. The authors report a minimum MAPE of 5.5682% for their proposed model.

Comparison

The present study addresses a similar business problem—understanding price variation in an online service marketplace—but uses a substantially simpler and more interpretable framework.

The two approaches differ in:

Dimension	Tan et al.	Present Study
Domain	Airbnb accommodation	Wedding photography
Data	8,380 listings	11,030 cleaned listings

Inputs	Structured + text + images	Structured vendor attributes
Main price model	Multimodal deep learning	Ridge hedonic benchmark
Secondary analysis	Price prediction	Fair-price classification
Market	Amsterdam	14 Indian markets

Tan et al. demonstrate the value of richer multimodal information for price prediction, while the present study prioritizes transparency and practical interpretability using structured public vendor attributes.

5.4 Research Gap

The reviewed studies demonstrate that online ratings, marketplace attributes and machine-learning methods can contribute to pricing analysis.

However, the present case study applies these ideas to a different context:

Indian wedding photography vendors.

The study combines:

1. Self-collected vendor-level web data
2. Market-specific price benchmarking
3. Out-of-fold expected-price estimation
4. Empirical residual-based pricing categories
5. Pre-booking classification

This creates a practical framework for assessing a vendor's quoted starting price relative to observable market characteristics.

The contribution should be viewed as a **domain-specific applied Business Analytics case study**, rather than as a claim of a completely new pricing theory.

6. Results, Insights and Recommendations

6.1 Key Findings

Finding 1 — Price variation is substantial

The median daily starting price was approximately ₹50,000, while the mean was approximately ₹66,312.

The difference reflects strong right-skewness in the market.

Business implication: A simple market-average price is unlikely to be sufficient for evaluating an individual vendor.

Finding 2 — Location is an important pricing signal

Prices differed considerably across the 14 markets.

The model also identified market effects as an important component of predictive classification.

Business implication: Customers should compare vendors primarily against relevant local markets rather than against a single national average.

Finding 3 — Review volume is an important predictive signal

Review count was the highest individual feature importance in the Random Forest model at approximately 23.6%.

Business implication: Marketplace reputation information may be useful when comparing vendors, particularly when star ratings are concentrated within a narrow range.

This does not establish that increasing review counts will automatically justify higher prices.

Finding 4 — Rating alone provides limited differentiation

Many rated vendors had high star ratings.

This concentration reduces the ability of star rating alone to distinguish vendors.

Business implication: Customers should consider rating together with review volume, location, service characteristics and other observable information rather than relying solely on the star score.

Finding 5 — Public listing information has limited predictive power

The OOF Ridge model achieved an R^2 of **0.0869**.

This means that most price variation remains unexplained by the variables available in the scraped listings.

This is an important limitation of the analysis.

Factors such as:

- exact coverage duration
- album specifications
- photographer availability
- artistic style
- negotiation
- destination travel
- seasonal demand
- client requirements

may influence actual pricing but were not sufficiently available in the structured dataset.

6.2 Recommendations for Customers

Customers can use the benchmark as an **initial screening tool**, rather than as a final purchasing decision.

Recommended approach:

1. Compare a quote against vendors in the same market.
2. Consider review volume along with rating.
3. Check actual package inclusions.
4. Examine whether higher prices correspond to additional deliverables.
5. Treat the model's pricing category as a benchmark signal rather than proof that a vendor is objectively overcharging.

A vendor classified as above the benchmark should therefore trigger further investigation rather than automatic rejection.

6.3 Recommendations for Photography Vendors

Vendors can use the benchmark to understand how their listed starting price compares with vendors having similar observable marketplace characteristics.

Useful actions include:

- reviewing pricing relative to the local market
- clearly communicating package inclusions
- maintaining accurate profile information
- transparently stating experience where appropriate
- monitoring review volume and customer feedback
- differentiating premium services clearly

The analysis does not establish a specific causal percentage by which a vendor should increase or decrease prices.

6.4 Recommendations for Marketplace Platforms

A wedding marketplace could use a benchmark similar to this study to improve price transparency.

Potential features include:

Fair-Price Guidance

Display an indicative expected-price range based on comparable vendors.

Comparable Vendor View

Allow customers to compare vendors within the same city and service category.

Explanation Layer

Instead of simply displaying "expensive" or "cheap", explain which observable factors contribute to the benchmark.

Emerging Vendor Support

Provide additional visibility to newer vendors whose lack of reviews may otherwise make them difficult to compare with established vendors.

These recommendations would require further testing using actual customer behaviour and booking outcomes before claiming improvements in conversion or booking performance.

6.5 Limitations

The study has several limitations.

1. Starting prices are not final negotiated prices

The dataset represents publicly displayed listing prices rather than finalized customer contracts.

2. Important service characteristics are unavailable

Information such as exact hours of coverage, album quality, editing style, equipment and photographer-specific artistic quality may not be consistently available.

3. Experience is frequently missing

Approximately 91.9% of vendor profiles do not explicitly state experience.

Median imputation was therefore used for continuous modelling, with an accompanying disclosure indicator.

4. Many vendors are unrated

Approximately 54.3% of vendors are unrated.

An unrated vendor may represent a new vendor, a vendor with limited platform activity or another form of missing marketplace information.

5. Observational data

The analysis identifies relationships and predictive patterns. It does not establish causal relationships.

6. Marketplace selection bias

The dataset represents vendors listed on the selected platform and may not represent every wedding photographer operating in India.

7. Temporal limitation

Prices and vendor profiles may change over time. The analysis represents the listings captured during the collection period.

8. Model performance

The Random Forest achieved 41.12% test accuracy. This indicates useful signal above random guessing but also demonstrates that observable listing attributes alone cannot reliably determine a vendor's pricing category in every case.

7. Conclusion

This case study developed an end-to-end Business Analytics framework for evaluating wedding photography vendor pricing using a self-collected web-scraped dataset.

A total of **11,340 raw vendor listings** were collected from publicly available WedMeGood listings across 14 Indian markets. After duplicate removal, missing-price filtering and extreme-value treatment, **11,030 records** remained for analysis.

The study first examined pricing patterns through exploratory analysis and found substantial geographic and distributional variation. Because price was strongly right-skewed, a logarithmic pricing specification was used.

A hedonic Ridge regression model was then implemented using **5-fold shuffled cross-validation** to generate out-of-fold expected prices. The resulting OOF residuals were used to construct empirical pricing categories. The lower and upper quartile boundaries were **-0.3363** and **+0.2973**, producing 25% Underpriced-relative-to-benchmark, 50% Fairly Priced-relative-to-benchmark and 25% Overpriced-relative-to-benchmark observations.

A second-stage classification model tested whether these categories could be predicted using information available before booking. Random Forest achieved **41.12% test accuracy and 0.3657 Macro F1**, compared with 39.17% accuracy for Logistic Regression and a 33.33% random baseline.

Review volume, market effects, rating and experience-related variables emerged as important predictive signals. However, the relatively modest predictive performance and the low OOF R^2 demonstrate that public listing attributes explain only a limited portion of wedding photography price variation.

Therefore, the primary value of the proposed framework is not to declare that a vendor is objectively expensive or inexpensive. Instead, it provides a **transparent benchmark-relative pricing signal** that can help customers, vendors and marketplace platforms make more informed comparisons.

Future work could incorporate richer information such as package duration, album specifications, photographer specialization, portfolio characteristics, seasonality, travel requirements and actual transaction prices. These additions could improve the ability of the model to distinguish genuine service-quality differences from simple marketplace price variation.

Overall, the study demonstrates how web-scraped business data, exploratory analytics, econometric benchmarking and machine learning can be combined to address a practical pricing-transparency problem in the wedding services marketplace.

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